

Question	Marking Guidance	Mark	Comments
07.1	1. Change in <u>DNA</u> base/nucleotide (sequence); 2. Change in amino acid (sequence)/primary structure; 3. Alters (position of) hydrogen/ionic/disulfide bonds; 4. Change in <u>tertiary</u> structure (of receptor);	4	1. Accept: mutation in <u>DNA</u> base (sequence). 1. Accept: deletion/substitution/addition of a <u>DNA</u> base/nucleotide. 2. Reject: different amino acid formed. 2. Ignore: change in code for amino acid. 4. Reject: any reference to active site. 4. Ignore: 3°.
07.2	1. (Receptor) is not <u>complementary</u> OR (HIV) cannot bind/attach <u>and</u> enter/infect (helper) T cell; 2. No replication (of virus) OR No destruction of (helper) T cell;	2	1. Accept: 'complimentary'. 1. Accept: invade as alternative to infect. 2. Accept: reproduction (of virus).
07.3	1. Low/lower exposure to HIV (in Europe) OR Low/lower number of HIV/AIDS (infections/cases); 2. (HIV) has only been present for a short time period OR (HIV relatively) recently evolved; 3. Mutation/CCR5 has been around for many years; 4. Mutation/CCR5 is advantageous (for something else);	2 max	1. Accept: converse. 3. Accept: frequency of mutation has always been high.